



Language-Agnostic Speech Tokenizer for Spoken Term Detection with Efficient Retrieval

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Abstract

The surge in multilingual and code-switched spoken content demands efficient Query-by-Example Spoken Term Detection (STD) systems capable of handling diverse languages. Existing STD systems are monolingual; they typically require large labeled datasets for training and use costly DTW-based matching during inference, limiting their practicality. This paper proposes a novel speech tokenizer that converts speech into language-agnostic tokens. Furthermore, a multi-stage search algorithm enables fast and efficient retrieval from large datasets. In experimental evaluations, the tokens from the proposed tokenizer demonstrate strong speaker invariance, consistent performance across languages, and a capability to generalize effectively to unseen languages, outperforming the baselines significantly.

Index Terms: Multilingual Speech Tokenization, Query-by-Example Spoken Term Detection, Audio Retrieval, Bidirectional Mamba, Optimal Transport

1. Introduction

Spoken Term Detection (STD) is the task of identifying instances of spoken queries within large audio collections, enabling applications like voice search, media monitoring, and spoken document retrieval. While STD systems have shown promising results in monolingual settings, the increasing prevalence of multilingual content across digital platforms necessitates robust multilingual STD solutions. This challenge is further amplified in code-switching scenarios, where speakers switch between languages within a single utterance, making accurate term detection significantly more complex yet essential for real-world applications.

ASR-based STD systems [1, 2] rely on ASR models to convert speech into phones or graphemes lattices. While effective in some settings, these systems face several limitations, especially in multilingual scenarios. First, they require large amounts of labeled data for each language, making it difficult to scale across multiple languages. Second, ASR systems often struggle with low-resource languages and code-switching, resulting in frequent transcription errors. Additionally, these systems are sensitive to pronunciation variations, accents, and domain-specific vocabulary, which can severely degrade their performance. Such challenges make ASR-based approaches less suitable for robust and scalable multilingual STD tasks.

Recently, a speech-tokenization-based approach [3] has emerged as a promising solution for STD tasks. Unlike ASR-based systems, which rely on transcriptions, the method learns speaker-agnostic tokens that capture phonetic and semantic content, enabling direct audio matching without intermediate text. Tokenization offers several key advantages: it enables fast search using text-based algorithms, allows compact storage of

large spoken archives, and supports compositionality, which is crucial for handling out-of-vocabulary (OOV) terms. This makes speech tokenization particularly suited for multilingual and code-switching scenarios, where ASR systems often struggle [1, 4].

In this paper, we propose a speech tokenization framework that generates language-agnostic and speaker-invariant speech tokens. The framework involves training a Bidirectional Mamba encoder [3] to learn contextual speech representations, which are subsequently discretized using a vector quantization (VQ) process [5]. However, the standard trainable codebook in VQ often suffers from the codebook collapse problem [6], leading to suboptimal token generation and degraded retrieval performance. To mitigate this, we adopt the approach from [7], which reformulates codebook learning as a balanced clustering problem using optimal transport [8]. This ensures uniform utilization of codewords and thus generates more balanced and informative speech tokens. For retrieval, we tokenize speech into discrete tokens and introduce a multistage search scheme that integrates text-based search algorithms, such as TF-IDF [9], to enhance spoken term retrieval efficiency.

We select a multilingual Indian speech database as a diverse and representative benchmark for evaluating our approach. Indian languages are spoken by a substantial portion of the global population; however, the development of multilingual speech technologies for these languages remains limited. Furthermore, Indian speech exhibits significant linguistic diversity, encompassing numerous languages, dialects, and frequent instances of code-switching. Additionally, we release a word-boundary annotated dataset for the Kathbath corpus [10], a 1500-hour speech dataset, which we use to train our model. Empirical results show that our proposed approach significantly outperforms baselines in both multilingual and monolingual settings and demonstrates generalization potential to unseen languages.

2. Proposed Method

Our goal is to generate consistent token sequences, $\hat{Z}$ and $\hat{Z}'$, for pairs of utterances (x, x') that correspond to the same spoken term w , regardless of the language, while ensuring that different terms yield distinct tokenized representations. The overall architecture of the proposed approach is shown in Figure 1.

2.1. Feature Encoder

We follow the architecture design for the Mamba feature encoder as outlined in [3]. The encoder $f : X \rightarrow Z$ consists of L layers of bidirectional Mamba blocks, which transform the input audio feature sequence $X = \{x_1, \dots, x_t\}$ into contextual speech embeddings $Z = \{z_1, \dots, z_t\}$. The final encoder layer

output is mapped to a d -dimensional space and is subsequently normalized using L_2 -normalization to ensure the sequence embeddings have a unit norm.

2.2. Contrastive Learning Framework

Speech Representation Learning. Our objective is to learn discriminative frame-level speech embeddings. To achieve this, we consider a pair of utterances (x, x') corresponding to a given spoken term w . To better reflect real-world scenarios—where utterances may contain additional context beyond the spoken term itself—we add contextual padding to each utterance, ensuring a fixed-length input. These padded utterances are then processed to extract their respective MFCC feature sequences, denoted as X and X' . Within the contrastive learning framework, we construct anchor-positive pairs by leveraging DTW-based alignment between X and X' . Notably, feature frames corresponding to padding are excluded during the alignment process, resulting in an alignment path:

$$\pi = \{(t, t') \mid t' = \arg \min_{i \in \mathcal{I}_t} d(X_t, X'_i), t \in [1, T], \mathcal{I}_t \subseteq [1, T']\}, \quad (1)$$

where $\mathcal{I}_t$ denotes the set of frame indices in X' aligned with X_t and $d(\cdot)$ is the Euclidean distance function.

Next, we feed X and X' to the feature encoder to obtain their corresponding embedding sequences Z and Z' . The alignment path π is then used for self-supervision to generate anchor-positive pairs as:

$$p_t = (z_t, z'_{t'}), \text{ where } (t, t') \in \pi \quad (2)$$

Hence, for each i^{th} training pair, we define the frame-level contrastive loss function as:

$$\mathcal{L}_{\text{contrastive}}^{(i)} = \frac{1}{T} \sum_{t=1}^T -\log \left(\frac{e^{(z_t \cdot z'_{t'})/\tau}}{e^{(z_t \cdot z'_{t'})/\tau} + \sum_{n=1}^N e^{(z_t \cdot z_n)/\tau}} \right), \quad (3)$$

where z_n are N negative embeddings randomly chosen from other pairs within a training batch, i.e., embeddings from terms $w' \neq w$.

Speech Tokenization. We use a VQ module to convert continuous embedding sequences Z and Z' into their corresponding discrete representations $\hat{Z}$ and $\hat{Z}'$. The VQ module employs a trainable codebook $\mathcal{C}$, which consists of K d -dimensional codewords, denoted as $\mathcal{C} = \{c_1, \dots, c_K\}$. In particular, VQ maps each frame-level embedding $z_t \in Z = \{z_1, \dots, z_T\}$ to its nearest codeword c_k in $\mathcal{C}$ based on cosine similarity as:

$$\hat{z}_t = \frac{c_{k^*}}{\|c_{k^*}\|_2}, \text{ where } k^* = \arg \max_{c_k \in \mathcal{C}} \left(z_t \cdot \frac{c_k}{\|c_k\|_2} \right), \quad (4)$$

Additionally, we introduce a commitment loss function to ensure that the frame-level embeddings closely align with their respective discrete representations as:

$$\mathcal{L}_{\text{commitment}}^{(i)} = -\frac{1}{T} \sum_{t=1}^T z_t \cdot \hat{z}_t + z'_t \cdot \hat{z}'_t \quad (5)$$

Robust Tokens. To ensure that frame-level embeddings within each aligned pair p_t are consistently mapped to the same codeword, we introduce an additional robustness loss function. This encourages the generation of robust and stable tokens, which

are crucial for effective retrieval. For the i^{th} training sample pair, the robustness loss is defined as:

$$\mathcal{L}_{\text{robustness}}^{(i)} = \frac{1}{|\pi|} \sum_{(t, t') \in \pi} \mathcal{L}(z_t, z'_{t'}) + \mathcal{L}(z'_{t'}, z_t) \quad (6)$$

where $\mathcal{L}(z_t, z'_{t'})$ denotes the cross-entropy loss between two distributions associated with z_t and $z'_{t'}$. This loss is computed as follows:

$$\mathcal{L}(z_t, z'_{t'}) = -\sum_{k=1}^K p(z_t|c_k) \log \left(\frac{\exp((z'_{t'} \cdot c_k)/\tau')}{\sum_{k'} \exp((z'_{t'} \cdot c_{k'})/\tau')} \right), \quad (7)$$

where $p(z_t|c_k)$ represents the probability of z_t being assigned to codeword c_k , and τ' is a parameter that controls the sharpness of the distribution. The reverse term $\mathcal{L}(z'_{t'}, z_t)$ is computed symmetrically to ensure bidirectional consistency.

Efficient Tokens. Typically, the probability $p(z_t|c_k)$ is modeled using a softmax function. However, this approach often leads to a skewed distribution due to the well-known index collapse problem during codebook training, where only a small subset of codewords is frequently selected while others remain underutilized. To address this, we adopt the optimal transport framework proposed in [7], which ensures a more uniform distribution of $p(z_t|c_k)$. This approach promotes balanced utilization of the codebook, effectively mitigating collapse and enabling the generation of more representative and diverse tokens.

Training objective. The model is trained using a total loss $\mathcal{L}$, which is computed over a training batch of size B . The total loss is formulated as:

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^B \mathcal{L}_{\text{contrastive}}^{(i)} + \alpha \mathcal{L}_{\text{commitment}}^{(i)} + \beta \mathcal{L}_{\text{robustness}}^{(i)}, \quad (8)$$

where α and β control the relative contribution of each loss component.

2.3. Retrieval

Given a set $\mathcal{A}$ of audio tracks, each track $a_i \in \mathcal{A}$ is divided into overlapping segments of length l_s with a hop size of h_s . For each segment, we compute its representation $Z = \{z_t\}_{t=1}^T$, followed by its corresponding tokenized representation $\mathcal{T} = \{i_t\}_{t=1}^T$, where each i_t denotes the index of the nearest codeword for z_t in the codebook $\mathcal{C}$. Next, we construct a TF-IDF representation for each tokenized sequence $\mathcal{T}$ in the database and index them using IVF-PQ [11] for efficient retrieval.

We perform retrieval in multiple stages, progressively filtering candidates at each stage to refine the results and improve precision. Given a query q , we first compute its token representation $\mathcal{T}_q$ and its corresponding TF-IDF representation. In the initial stage, we retrieve a set of candidate matches $\mathcal{P}_1$ using the index. In the second stage, we further refine $\mathcal{P}_1$ by computing the Jaccard similarity [12] between $\mathcal{T}_q$ and $\mathcal{T} \in \mathcal{P}_1$, resulting in a filtered set $\mathcal{P}_2$. Lastly, we apply an edit distance-based filtering on $\mathcal{P}_2$ to obtain the final set of best-matches $\mathcal{P}_3$. The edit distance is used to incorporate temporal information, which is absent in the Jaccard similarity computation performed in the previous stage. Our retrieval process is limited to this stage

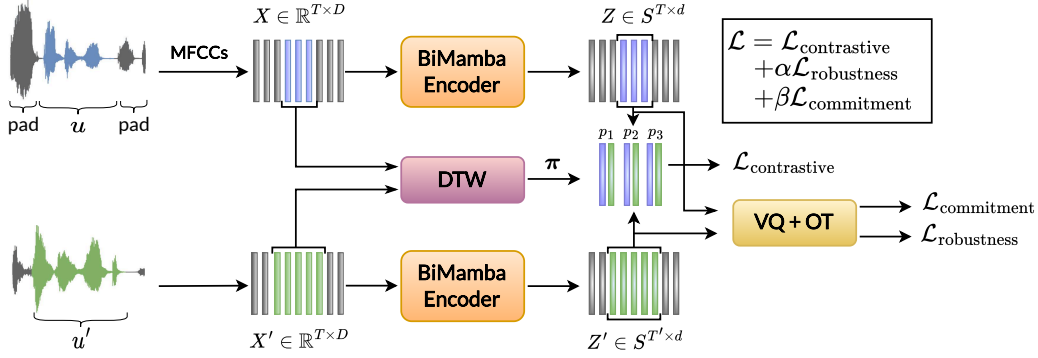


Figure 1: Training pipeline of our speech tokenization model.

for efficiency reasons. However, to further improve precision, DTW can be applied as additional filters using the discrete representation $\hat{Z}$ and the continuous representation Z in succession.

3. Experiments

3.1. Databases

We used the Kathbath database [10], a multilingual Indian speech dataset containing read speech. This dataset is highly diverse, encompassing multiple dialects, languages, and speakers. Additionally, it includes noisy speech and instances where English words are interspersed within regional language utterances. To obtain word-level alignments for the corpus, we used the Montreal Forced Aligner [13]. Our model was trained on a subset of the *train* split using these alignments.¹

3.2. Evaluation

We conducted our evaluation using the test splits of the dataset. Two distinct query sets, each containing 2,500 spoken queries, were created. Each query corresponds to a term with at least five characters in its text form. The query sets are defined as follows:

- **Test Set 1:** This set is derived from the *test_known* split of the dataset. It contains terms spoken by speakers who were seen during training.
- **Test Set 2:** This set is constructed from the *test_unknown* split. It includes terms spoken by speakers who were not encountered during training.

We evaluated the baselines using Maximum Term Weighted Value (MTWV) [14] and Recall as performance metrics.

3.3. Baselines

We evaluated our approach against token-based STD baselines, where tokens were either posterior tokens obtained from an ASR model or KMeans-based tokens derived from intermediate layer representations. Given the unavailability of a multilingual ASR model trained on Indian languages, we opted for monolingual HuBERT [15] models to extract both types of tokens for each language under evaluation.

¹Our word-level alignments for entire corpus are available for download at <https://zenodo.org/records/1550222>

Table 1: STD results in the monolingual settings.

Language	Model	Test set 1 (<i>test_known</i>)		Test set 2 (<i>test</i>)	
		MTWV $\uparrow$	Recall $\uparrow$	MTWV $\uparrow$	Recall $\uparrow$
Hindi	HuBERT-KMeans	0.58	0.63	0.57	0.63
	HuBERT-Posteriors	0.60	0.75	0.61	0.75
	Ours-Transformer	0.78	0.86	0.77	0.87
	Ours-Bimamba	0.79	0.87	0.77	0.87
Marathi	HuBERT-KMeans	0.39	0.41	0.44	0.47
	HuBERT-Posteriors	0.58	0.76	0.54	0.78
	Ours-Transformer	0.74	0.85	0.72	0.85
	Ours-Bimamba	0.75	0.86	0.71	0.84
Punjabi	HuBERT-KMeans	0.52	0.55	0.49	0.52
	HuBERT-Posteriors	0.55	0.66	0.56	0.68
	Ours-Transformer	0.75	0.85	0.76	0.85
	Ours-Bimamba	0.76	0.85	0.75	0.84
Gujarati	HuBERT-KMeans	0.60	0.67	0.55	0.61
	HuBERT-Posteriors	0.59	0.74	0.58	0.76
	Ours-Transformer	0.79	0.89	0.78	0.89
	Ours-Bimamba	0.80	0.89	0.78	0.89
Sanskrit	HuBERT-KMeans	0.48	0.68	0.43	0.63
	HuBERT-Posteriors	0.56	0.84	0.55	0.81
	Ours-Transformer	0.69	0.87	0.69	0.88
	Ours-Bimamba	0.70	0.87	0.70	0.88
Odia	HuBERT-KMeans	0.47	0.53	0.46	0.51
	HuBERT-Posteriors	0.58	0.77	0.54	0.77
	Ours-Transformer	0.61	0.68	0.58	0.68
	Ours-Bimamba	0.61	0.69	0.58	0.70
Tamil	HuBERT-KMeans	0.45	0.50	0.37	0.41
	HuBERT-Posteriors	0.44	0.82	0.43	0.87
	Ours-Transformer	0.62	0.84	0.59	0.85
	Ours-Bimamba	0.62	0.85	0.60	0.84
Telugu	HuBERT-KMeans	0.52	0.57	0.50	0.60
	HuBERT-Posteriors	0.63	0.83	0.59	0.82
	Ours-Transformer	0.73	0.84	0.70	0.83
	Ours-Bimamba	0.72	0.84	0.70	0.83
Kannada	HuBERT-KMeans	0.48	0.56	0.46	0.52
	HuBERT-Posteriors	0.54	0.77	0.52	0.78
	Ours-Transformer	0.73	0.85	0.71	0.86
	Ours-Bimamba	0.73	0.86	0.72	0.85
Bengali (unseen to ours)	HuBERT-KMeans	0.53	0.58	0.57	0.60
	HuBERT-Posteriors	0.60	0.81	0.55	0.80
	Ours-Transformer	0.70	0.80	0.68	0.79
	Ours-Bimamba	0.71	0.79	0.70	0.78

3.4. Implementation

The input audio segments were fixed at 1 s, capturing approximately 95% of spoken terms across different languages in the dataset. This choice ensured comprehensive term coverage during training. The encoder architecture comprised 16 bidirectional Mamba layers, followed by a projection layer that mapped the output to a 32-dimensional embedding space, resulting in a total of 16.7M trainable parameters. The hyperparameter τ and τ' were set to 0.1, while α and β were assigned values of 1 and 10, respectively. The model was trained for 650k steps using the Adam optimizer with a learning rate of 5×10^{-4} . For retrieval, we statistically determined filter thresholds at different stages of the search for each method.

Table 2: Average MTWV ($\uparrow$) and Recall ($\uparrow$) scores for different methods in multilingual settings..

Model	Hindi		Marathi		Punjabi		Gujarati		Tamil		Telugu		Sanskrit	
	MTWV	Recall	MTWV	Recall	MTWV	Recall	MTWV	Recall	MTWV	Recall	MTWV	Recall	MTWV	Recall
HuBERT-Posteriors:														
Hindi	0.40	0.42	0.35	0.36	0.35	0.36	0.34	0.35	0.35	0.37	0.34	0.36	0.37	0.38
Marathi	0.34	0.35	0.34	0.35	0.33	0.34	0.41	0.43	0.33	0.34	0.32	0.33	0.39	0.41
Punjabi	0.29	0.30	0.26	0.27	0.27	0.28	0.24	0.25	0.27	0.28	0.27	0.28	0.29	0.30
Gujarati	0.46	0.48	0.39	0.41	0.37	0.39	0.43	0.44	0.38	0.40	0.37	0.38	0.40	0.41
Tamil	0.30	0.33	0.34	0.37	0.32	0.36	0.26	0.30	0.37	0.41	0.37	0.40	0.34	0.38
Telugu	0.43	0.45	0.44	0.46	0.42	0.43	0.38	0.40	0.44	0.46	0.46	0.48	0.46	0.47
Sanskrit	0.43	0.45	0.42	0.45	0.41	0.44	0.39	0.41	0.44	0.47	0.45	0.47	0.46	0.48
HuBERT-KMeans:														
train lang same as query lang	0.50	0.51	0.31	0.32	0.34	0.35	0.44	0.46	0.43	0.46	0.41	0.42	0.46	0.47
Ours-Transformer	0.64	0.67	0.66	0.68	0.70	0.71	0.66	0.67	0.56	0.60	0.60	0.63	0.67	0.69
Ours-Bimamba	0.67	0.71	0.65	0.69	0.68	0.72	0.69	0.71	0.61	0.64	0.59	0.62	0.70	0.73

4. Results and Discussion

4.1. Monolingual Retrieval

We first evaluate our model performance in monolingual settings, where the retrieval database and query belong to the same language. The results, summarized in Table 1, demonstrate that our approach consistently outperforms baseline methods. Despite being significantly smaller in size and trained on a much smaller dataset than the HuBERT model, our method achieves superior retrieval performance. Notably, our model delivers similar performance on both query type sets, highlighting its robustness to unseen speaker identities. It is crucial to note that our model demonstrates reliable performance on Bengali, a language not seen during training. This suggests that our method generalizes well to unseen and low-resource languages.

4.2. Multilingual Retrieval

We further evaluate our method in a multilingual retrieval setting, where both the retrieval database and queries comprise multiple languages. Specifically, we use the *test* split of the database to assess our model performance on unseen speaker queries, ensuring robustness to speaker variability. To evaluate the cross-lingual performance of the ASR-based approaches, we select a monolingual ASR model, extract posterior tokens from the multilingual database and queries, and measure retrieval performance. The results for this setup are presented in the first row of Table 2. For the baseline method using KMeans-based tokens, we train a KMeans model using features extracted from audio in the same language as the monolingual ASR model. This trained KMeans model is then used to extract tokens for the multilingual database and evaluate retrieval performance, as shown in the second row of the Table 2. In contrast, our multilingual model directly tokenizes the entire retrieval database.

Our results show that our method consistently outperforms baseline methods across multiple languages. This superior performance can be attributed to the efficient, language-agnostic tokens generated by our approach, which effectively capture phonetic information and leverage similarities in speech sounds across languages. As a result, speech segments with shared characteristics are mapped to consistent tokenized representations, enabling robust retrieval in diverse linguistic settings. Note that, our transformer-based model, trained within the same learning framework, performs competitively compared to the bidirectional Mamba model.

We observe a decline in performance for both proposed methods and the baseline approaches in the multilingual setting compared to the monolingual scenario. This drop can be

attributed to several factors. First, the retrieval database in the multilingual setting is substantially larger, encompassing all language databases, which increases the search space and complexity. Additionally, the presence of homophones within individual languages introduces ambiguity, leading to further performance degradation. Another contributing factor is the existence of words that appear across multiple languages, which can result in unintended matches. Importantly, while these cases may be classified as errors in the evaluation metrics, they are not necessarily true errors from the perspective of direct audio matching. Since our approach operates at the audio level rather than relying on linguistic distinctions, these instances reflect the inherent nature of spoken language rather than deficiencies in the retrieval system. However, because standard evaluation metrics do not account for such nuances, they register as performance drops in our results.

Notably, the HuBERT model, utilizing KMeans-based tokens, demonstrates superior performance compared to its corresponding posterior tokens when evaluated on queries in the same language as the model’s training data. This suggests that the discretization process introduced by KMeans contributes to more robust token representations. Furthermore, these findings highlight the potential for further investigation into the utility of intermediate-layer features, particularly in cross-lingual settings.

5. Conclusion

This paper introduces a speech tokenization method designed to generate efficient, language-agnostic tokens for multilingual spoken content retrieval. The proposed tokens effectively capture semantic information while remaining independent of language constraints. This capability is particularly significant for low-resource languages, as demonstrated by the model’s strong performance on an unseen language. Additionally, to enhance retrieval efficiency, we employ a multistage search strategy that integrates TF-IDF, facilitating fast and accurate spoken term detection. The experiments demonstrate that the proposed approach significantly outperforms monolingual ASR-based STD baselines while maintaining high retrieval efficiency. These findings underscore the potential of our approach in improving multilingual speech retrieval, particularly in resource-constrained settings. Our code is available at <https://github.com/anupsingh15/LAST>

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